



PRAHADEESH NARENDRAN THIMMA

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B.Tech. Electronics Engineering (VLSI Design and Technology)

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Aspiring VLSI Engineer with strong foundations in digital logic, Verilog-based design, and hardware debugging. Developing skills in design verification and simulation to ensure functional correctness and reliable digital circuit performance. Also experienced in embedded systems development.

EXPERIENCE

- Team RoboManipal

Team Lead, Electronics Engineer

May 2023 - Present

Manipal, India

 - Achieved 1st Place in Ather Energy’s national-level firmware engineering competition, demonstrating exceptional problem-solving and embedded debugging skills.
 - Developed C-based firmware on STM32 for Waveshare ST3215 and Dynamixel AX-series servos, implementing precise real-time control via half duplex UART.
 - Successfully completed consultancy work for Project Dhoopnavarti, delivering a custom PCB design and firmware tailored to the client’s requirements.
 - Winners at Hardware Hackathon 2.0, built an actuator fault detector to monitor health of any electromagnetic actuator using parameters like voltage, current, vibration and temperature.
 - Wrote firmware for hockey-playing robots using ESP32 for high-speed processing and Bluetooth for wireless control.
 - Leading the interdisciplinary team, managing projects, defining technical roadmaps, and mentoring members across technical domains.

- ChipMIT

Core Committee Member

May 2025 - Present

Manipal, India

 - Collaborated on digital design and verification projects using Verilog, focusing on RTL design, simulation, and testbench validation.
 - Mentored peers in understanding protocol design, synthesis flow, and debugging logic-level issues.

Projects:

 - I²C on FPGA:** Designed and implemented an I²C master controller in Verilog to interface with the Nexys A7 on-board temperature sensor. Generated a 10 kHz SCL from a 200 kHz clock, implemented a 28 state FSM for start condition, address transmission, ACK/NACK handling, and MSB/LSB temperature data reception. Verified correct I²C timing and temperature data capture through simulation.
 - 8-bit Low Power ALU Design in Verilog:** Designed and tested an optimized low-power ALU using Carry Skip Adder, Vedic Multiplier, and Non-Restoring Divider. Implemented power-saving techniques and verified arithmetic functionality and timing behavior through simulation and waveform analysis.

PROJECTS

- PWM Generator in Verilog

 - Designed and implemented a 16-bit PWM module emulating STM32 timers with prescaler (PSC), auto-reload (ARR), and compare (CCR) registers.
 - Implemented shadow-to-active register transfer to mimic STM32 preload behavior, ensuring glitch-free PWM updates.
 - Verified PWM output generation in simulation, including configurable duty cycle and frequency, and explained timing behavior using update events (UEV).
- Firmware Development for Waveshare and Dynamixel Servos

 - Developed and validated STM32-based C firmware for high-speed half duplex UART communication to control multiple servos in a closed-loop setup.
 - Debugged bus communication and verified real-time response using a logic analyzer and serial waveform inspection.
- SAD (Sum of Absolute Differences) Algorithm Implementation in Cadence

 - Implemented the SAD algorithm at RTL level in Cadence for block-matching-based image processing.
 - Designed modular Verilog components and simulated them to verify correct pixel-level difference and accumulation behavior.
 - Performed synthesis using Cadence Genus and analyzed physical design reports for power, performance, and area metrics.

SKILLS

- Languages:** C, Python, Verilog, SystemVerilog
- Tools:** STM32CubeIDE, Vivado, Cadence Genus, Cadence Virtuoso, Sentaurus TCAD
- Hardware Tools:** Logic Analyzer, Oscilloscope, Serial Monitor
- Firmware Tools:** STM32 HAL, Keil, VSCode
- Soft Skills:** Team work, Problem Solving, Documentation, Team Leadership

EDUCATION

Degree/Certificate	Institute/Board	CGPA/%	Year
B.Tech. EE (VLSI Design and Tech.)	Manipal Institute Of Technology, Manipal	7.65	2023–2027
Senior Secondary	CBSE Board	87.7%	2023
Secondary	CBSE Board	83.6%	2021